

Week 6, October 7, 2026

Main topics

- Generalized Tikhonov and Total Variation.
- Advanced Regularization Techniques.

Material

From last week: P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, Chapter 8. The slides are in the file Chap8.pdf.

New material this week: Statistical and Computational Inverse Problems by Jari Kaipio and Erkki Somersalo, sections 3-3.2.1 (pp 49-58) and 3.4 (pp 72-79).

Program

13:00-13:30 Generalized Tikhonov and Total Variation – short lecture.

13:30-15:30 Generalized Tikhonov and Total Variation – exercises (see below).

15:45-17:00 Bayesian Inverse Problems and Tikhonov.

Exercise 1 – First order derivative matrices

In the lecture, we introduce the matrix L , which represent the first order derivative operator, based on differences between neighbouring elements of the solution vector x , representing samples of the solution at the grid points $t_j = (j - \frac{1}{2})/n$, $j = 1, \dots, n$. Alternatively, we can use central differences that involve elements with doubled distance, i.e., x_{j-1} and x_{j+1} . Show that the corresponding matrix for the first derivative are given by

$$\hat{L}_1 = \frac{1}{2} \begin{pmatrix} -1 & 0 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 0 & 1 \end{pmatrix}.$$

Also show that the null spaces of this matrix is

$$\mathcal{N}(\hat{L}_1) = \text{span}\{(1, 1, \dots, 1)^T, (1, -1, 1, -1, \dots)^T\}.$$

Why is it dangerous to use $\hat{L}_1$ in the regularization term?

Exercise 2 – Generalized Tikhonov regularization

This exercise illustrates that Tikhonov regularization, corresponding to $L = I$, is not always a good choice in regularization problems, and the solution reflects the prior information that is

cooperated through the regularization. The test problem here is Abel integral problem, which should have been implemented in Week 2.

- Add Gaussian white noise to the right-hand side b . For example, with standard deviation $\eta = 0.01$.
- Compute ordinary Tikhonov solutions and vary the regularization parameter λ until the noise starts to dominate x_λ , and plot the solutions. Find the optimal solution according to the relative error.
- Generate the first derivative smoothing matrix

$$L_1 = \begin{pmatrix} -1 & 1 & & \\ & \ddots & \ddots & \\ & & -1 & 1 \end{pmatrix} \in \mathbb{R}^{(n-1) \times n}. \quad (1)$$

Compute the generalized Tikhonov solutions with L_1 . Again, vary λ until the noise starts to dominate $x_{L,\lambda}$. Find and plot the optimal solution according to the relative error.

- Repeat the previous step with the second derivative matrix

$$L_2 = \begin{pmatrix} 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \end{pmatrix} \in \mathbb{R}^{(n-2) \times n}.$$

- Compare all the optimal solutions from different regularization terms. What are the main differences? Do the solutions fit to the corresponding prior information?
- If you have time, you can try your test on the other two model problems: Phillips and backwards heat equation to see if you get the same conclusions.

Exercise 3 – Total variation regularization

In this exercise, we use the same test problem as in Exercise 2 to illustrate the performance of the total variation regularization. The implementation of the optimization solver for

$$\min_x \frac{1}{2} \|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda \text{TV}(\mathbf{x})$$

can be found in DTU Learn, where you need input A , L_1 defined in (1), $\mathbf{b}$ and λ .

- Vary the regularization parameter λ and observe how the solution $\mathbf{x}_\lambda$ changes.
- Find the optimal solution according to the relative error, and plot it.
- Compare the optimal solution from TV regularization with the optimal solutions from Exercise 2. What are the main differences?